_eda_price_2025_Q4

March 22, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 34,014 quotes from 23,888 customers
```

```python
#customers = aggregate_customer(df_clean)
```

```python
all_customers = pd.read_csv('customer_master_data.csv')
```

```python
# Using quarter column (if you have it)
customers_2025_q4 = all_customers[
    (all_customers['year'] == 2025) &
    (all_customers['quarter'] == 4)
]

# Using date range
customers_2025_q4 = all_customers[
    (all_customers['first_quote_date'] >= '2025-10-01') &
    (all_customers['first_quote_date'] <= '2025-12-31')
]

print(f"Q4 2025 customers: {len(customers_2025_q4):,}")
```

```
Q4 2025 customers: 2,867
```

```python
customers = customers_2025_q4
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
```

```
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')

visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.305729            1920            587.0
True                      0.407603             947            386.0

Chi-square p-value: 7.604283e-08
Statistically significant: YES

Conversion without issues: 30.57%
Conversion with issues: 40.76%
Difference: 10.2 percentage points
```

```
Customer-level finding: +10.2 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.32
  With issues: 1.48
  Ratio: 1.12x

avg_quote_amount:
  No issues: 5662.83
  With issues: 10339.86
  Ratio: 1.83x

max_quote_amount:
  No issues: 5967.72
  With issues: 11068.71
  Ratio: 1.85x

customer_duration_days:
  No issues: 2.47
  With issues: 4.04
  Ratio: 1.63x

price_range:
  No issues: 674.05
  With issues: 1387.49
  Ratio: 2.06x

multiple_quotes:
  No issues: 0.25
  With issues: 0.34
  Ratio: 1.36x

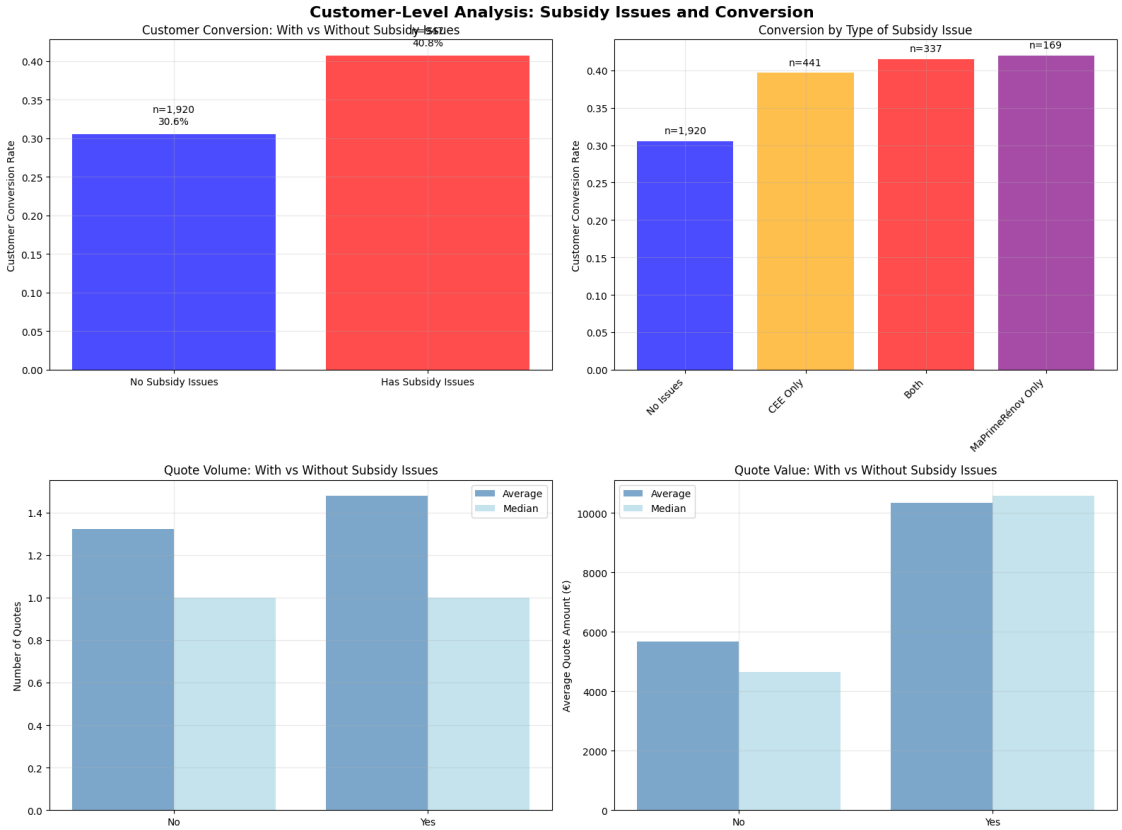
Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.748  0.657
Three+ quotes      0.051  0.090
Two quotes         0.201  0.253

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
```

```
Long (>30 days)     0.018  0.041
Medium (8-30 days)  0.034  0.056
Quick (2-7 days)    0.041  0.070
Same day decision   0.907  0.833

Product family by subsidy issue status:
had_subsidy_issue                 False  True
main_product_family
Appareil hybride                  0.005  0.046
Autres                            0.010  0.002
Chaudière                         0.476  0.057
Climatisation                     0.214  0.088
ECS : Chauffe-eau ou adoucisseur  0.058  0.017
Emetteur de chauffage  ou chappe  0.004  0.004
Photovoltaïque                    0.012  0.011
Plomberie Sanitaire               0.004  0.000
Pompe à chaleur                   0.041  0.477
Poêle                             0.168  0.298
Produit VMC                       0.006  0.000
Unknown                           0.002  0.000

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
MaPrimeRénov Only   0.420118    169
Both                0.415430    337
CEE Only            0.396825    441
No Issues           0.305729   1920
```

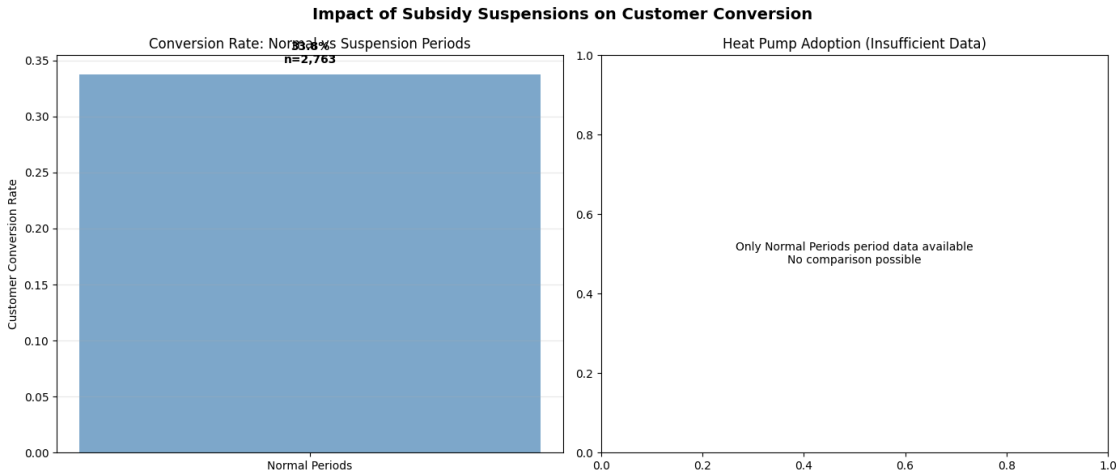



```
================================================================================
Policy Suspension Impact
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
Unique values in during_suspension: [False]

   Only data during normal periods available

Conversion during subsidy suspensions:
                    mean  count
Normal Periods  0.337676   2763
```

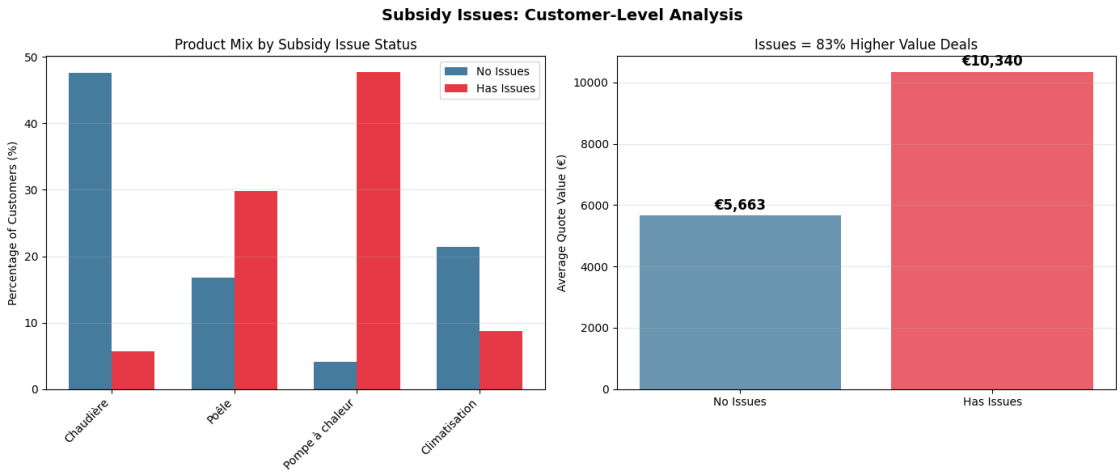



```
Unique values in main_product_family:
main_product_family
Chaudière                           968
Poêle                               604
Pompe à chaleur                     531
Climatisation                       493
ECS : Chauffe-eau ou adoucisseur    128
Appareil hybride                     54
Photovoltaïque                       34
Autres                               21
Produit VMC                          12
Emetteur de chauffage  ou chappe     11
Plomberie Sanitaire                   8
Unknown                               3
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                           968
Poêle                               604
Pompe à chaleur                     531
Climatisation                       493
ECS : Chauffe-eau ou adoucisseur    128
Appareil hybride                     54
Photovoltaïque                       34
Autres                               21
Produit VMC                          12
Emetteur de chauffage  ou chappe     11
Name: count, dtype: int64

Using these products: ['Chaudière', 'Poêle', 'Pompe à chaleur', 'Climatisation']
```

```
No Issue %: [np.float64(47.604166666666664), np.float64(16.770833333333332),
np.float64(4.114583333333333), np.float64(21.354166666666664)]
Has Issue %: [np.float64(5.70221752903907), np.float64(29.778247096092926),
np.float64(47.72967265047519), np.float64(8.764519535374868)]
```

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================================================================================
Threshold effect
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Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
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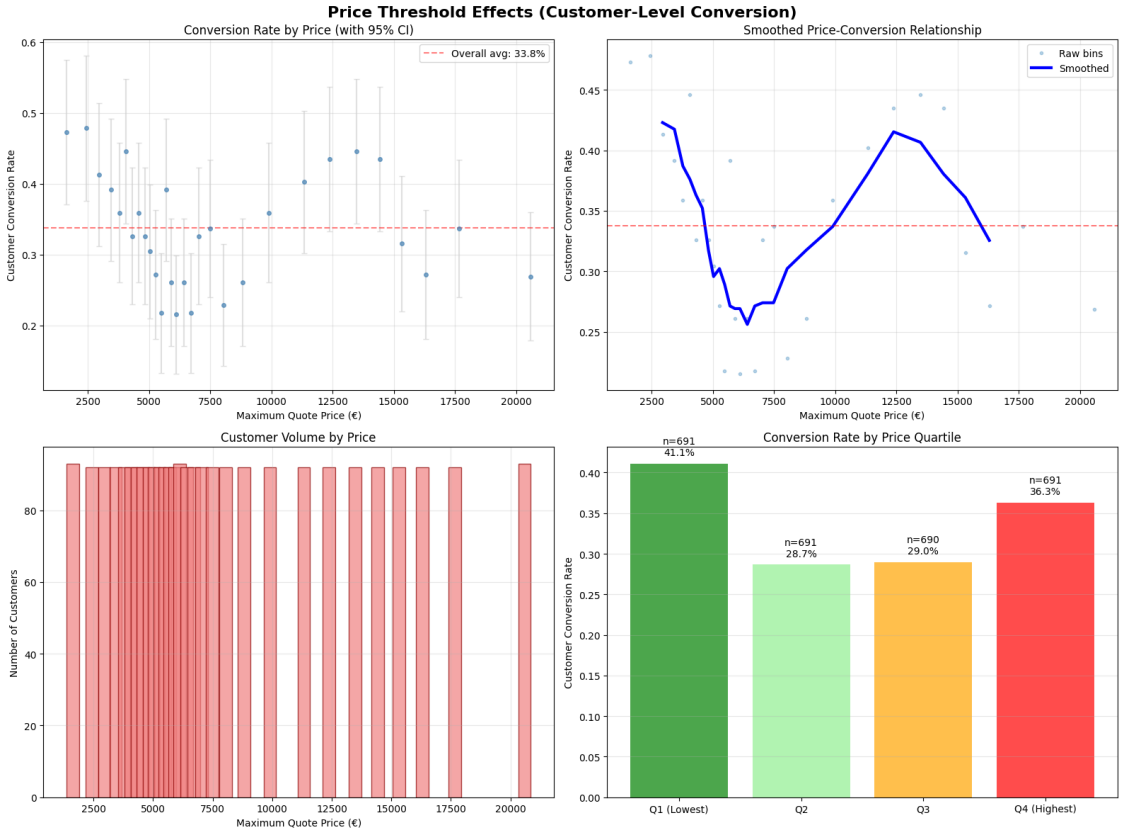


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================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 41.1% (n=691)
  Q2: 28.7% (n=691)
  Q3: 29.0% (n=690)
  Q4 (Highest): 36.3% (n=691)

Difference between lowest and highest quartile: 4.8 percentage points

Sweet spots (above-average conversion):
  €1,630: 47.3% (n=93)
  €2,441: 47.8% (n=92)
  €2,952: 41.3% (n=92)
  €3,435: 39.1% (n=92)
  €3,781: 35.9% (n=92)
  €4,067: 44.6% (n=92)
  €4,577: 35.9% (n=92)
```

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  €5,706: 39.1% (n=92)
  €9,902: 35.9% (n=92)
  €11,326: 40.2% (n=92)
  €12,390: 43.5% (n=92)
  €13,489: 44.6% (n=92)
  €14,428: 43.5% (n=92)

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
price_quartile
Q1_Low                   21.8    47.7        0.3   20.4    9.5      0.3
Q2_MidLow                14.2    40.6        0.3    1.6   43.3      0.1
Q3_MidHigh               22.8    37.4        8.1    2.7   28.9      0.0
Q4_High                  12.8    10.1       73.0    3.3    0.9      0.0

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=597):
  Q1_Low: 0.0%
  Q2_MidLow: 50.0%
  Q3_MidHigh: 57.9%
  Q4_High: 41.0%

Boiler (n=959):
  Q1_Low: 46.3%
  Q2_MidLow: 29.4%
  Q3_MidHigh: 31.1%
  Q4_High: 26.8%

AC (n=507):
  Q1_Low: 33.8%
  Q2_MidLow: 28.0%
  Q3_MidHigh: 24.2%
  Q4_High: 18.9%

Stove (n=604):
  Q1_Low: 34.3%
  Q2_MidLow: 29.2%
```

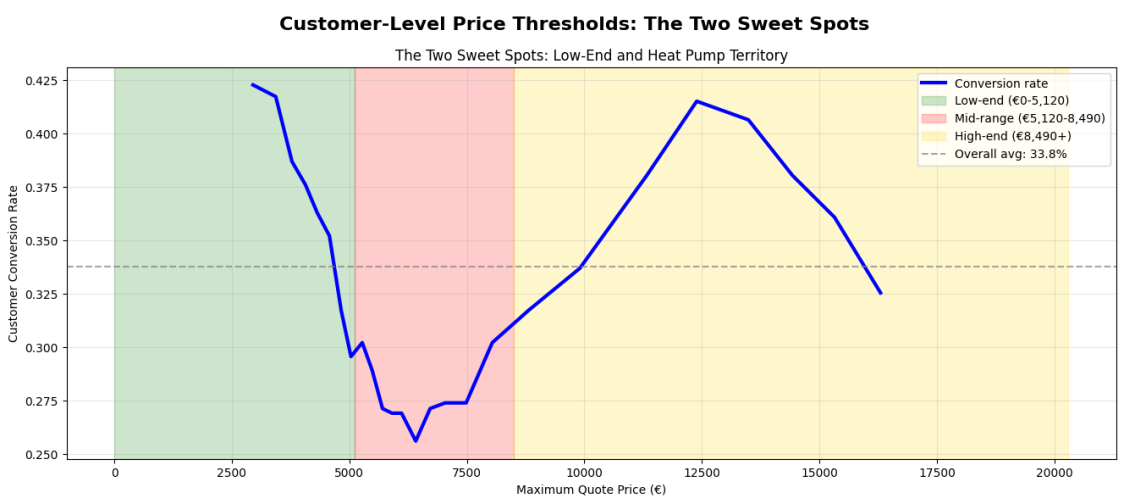
```
  Q3_MidHigh: 22.5%
  Q4_High: 66.7%

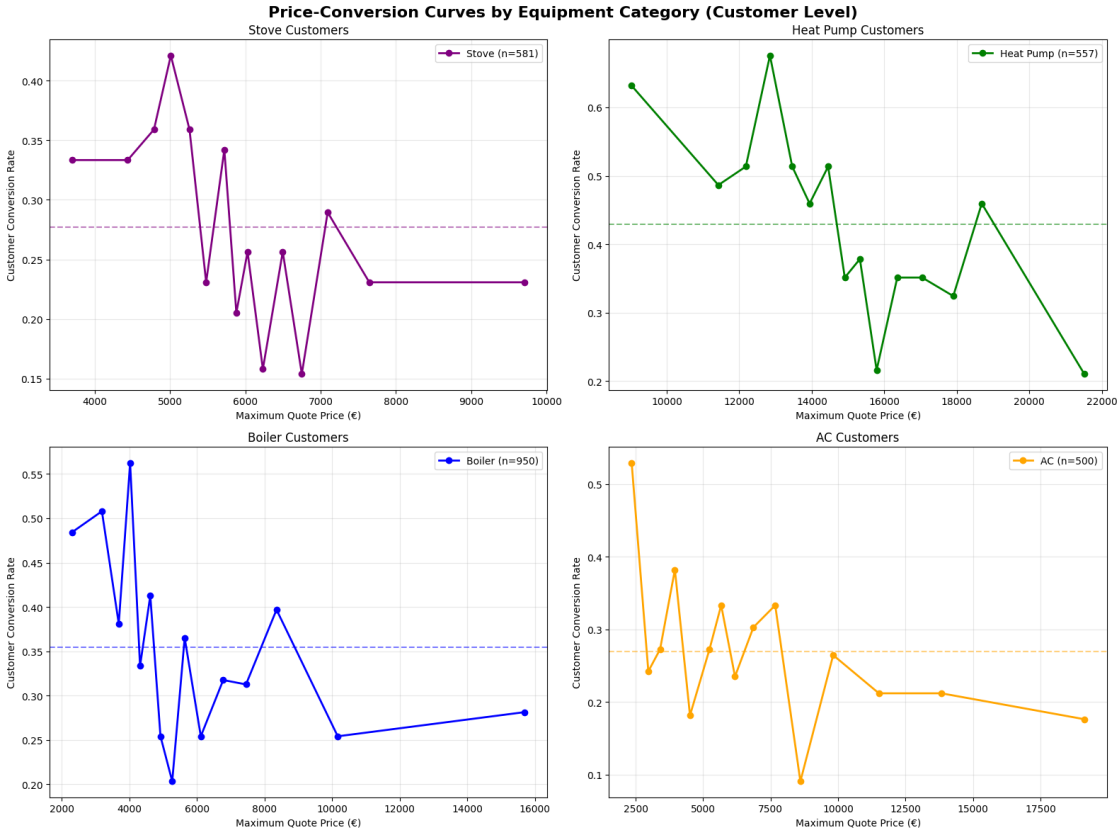
================================================================================
HEAT PUMP CUSTOMERS (n=674)
================================================================================
Average price: €15,634
Conversion rate: 40.8%
Subsidy issue rate: 84.3%

Heat pump adoption by quartile:
  Q1_Low: 0.4%
  Q2_MidLow: 0.4%
  Q3_MidHigh: 9.5%
  Q4_High: 82.3%

================================================================================
Equipment Category Segment Effect
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
```



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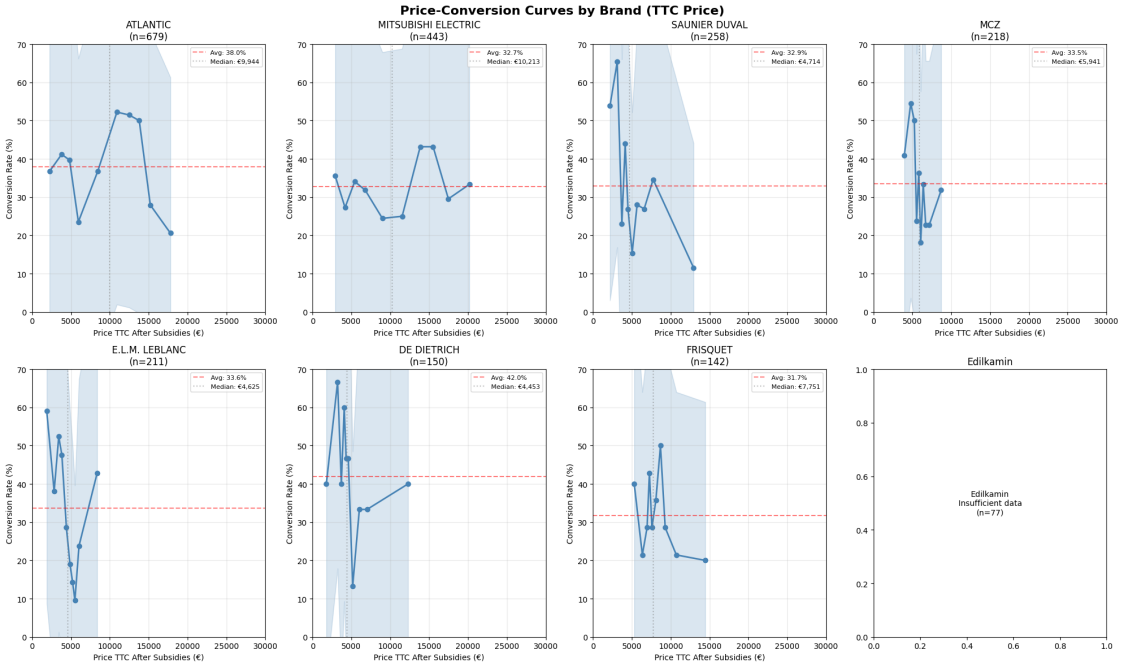
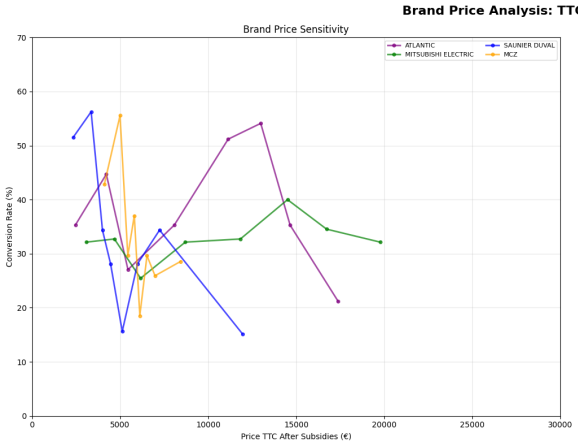
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'SAUNIER DUVAL', 'MCZ',
'E.L.M. LEBLANC', 'DE DIETRICH', 'FRISQUET', 'Edilkamin']
```




```
Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
```



Brand Performance Summary

Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	679	38.0%	€9,541	€11,851-14,486	50.7%
MITSUBISHI ELECTRIC	443	32.7%	€10,681	€12,693-16,460	43.2%
SAUNIER DUVAL	258	32.9%	€5,567	€1,343-3,421	59.6%
MCZ	218	33.5%	€6,052	€1,800-5,063	47.7%
E.L.M. LEBLANC	211	33.6%	€4,620	€3,261-4,098	50.0%
DE DIETRICH	150	42.0%	€5,215	€1,371-3,559	53.3%
FRISQUET	142	31.7%	€8,487	€8,474-9,725	39.3%

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================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 2,763 customers
```

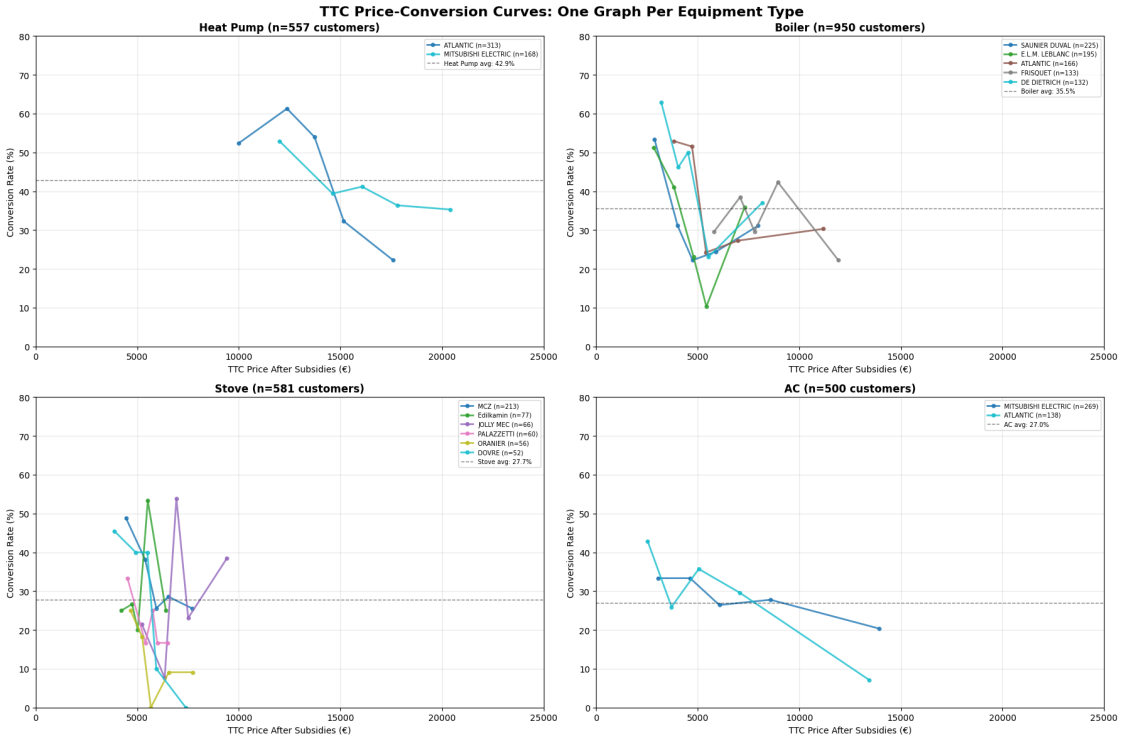
```
Price range: €1,343 to €25,369
Median price: €6,251

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC']

Boiler - Top brands: ['SAUNIER DUVAL', 'E.L.M. LEBLANC', 'ATLANTIC', 'FRISQUET',
'DE DIETRICH']

Stove - Top brands: ['MCZ', 'Edilkamin', 'JOLLY MEC', 'PALAZZETTI', 'ORANIER',
'DOVRE']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC']
```

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================================================================================
Seasonal Effects on Customer Conversion
================================================================================
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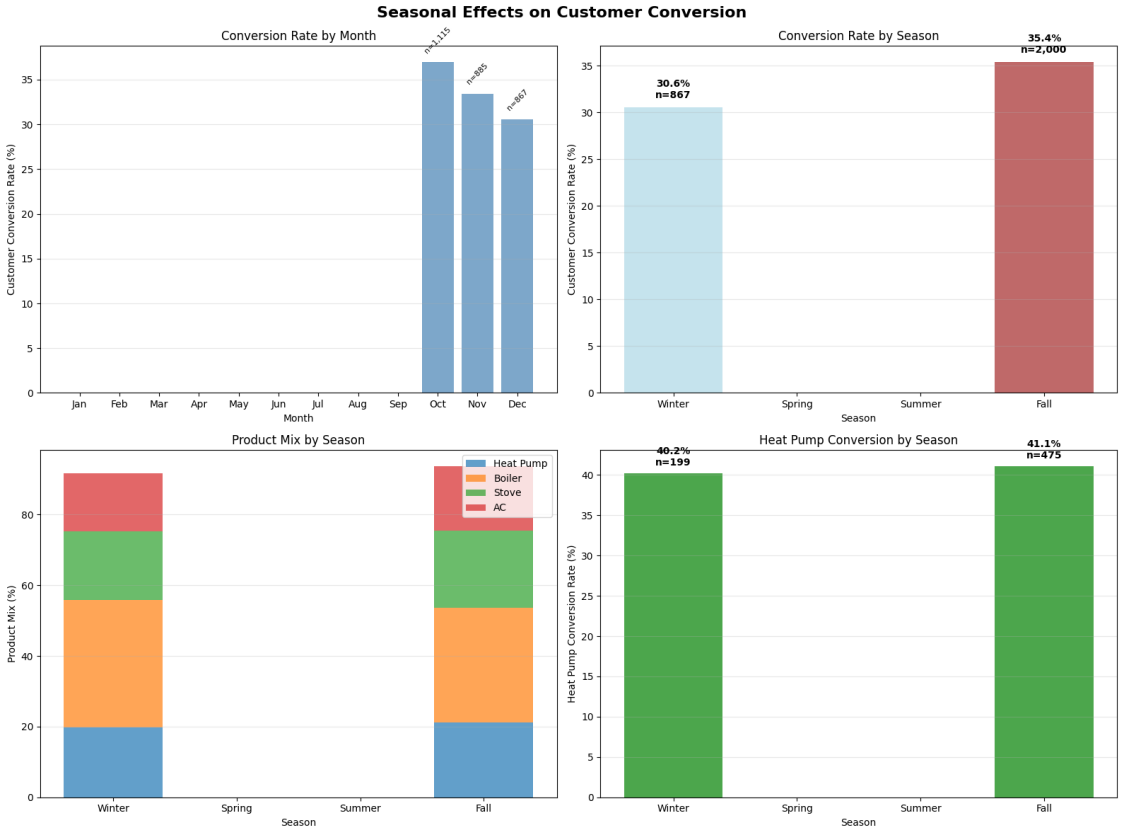



```
================================================================================
SEASONAL STATISTICS
================================================================================

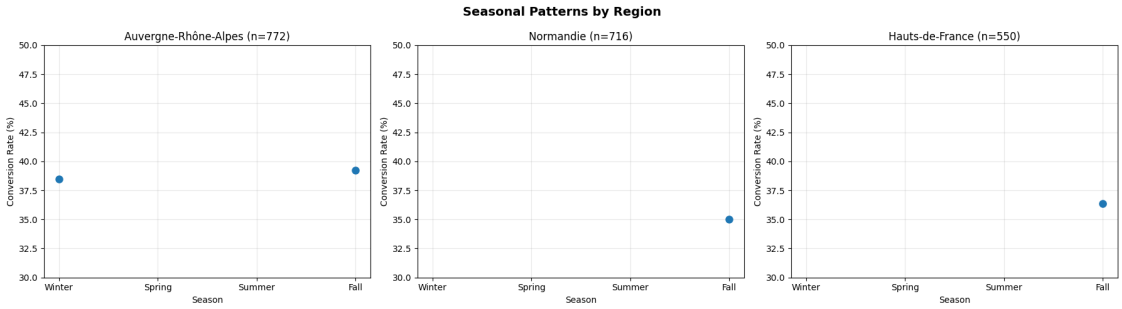
Conversion by season:
  Winter: 30.6% (n=867)
  Fall: 35.4% (n=2,000)

Seasonality chi-square p-value: 0.0136
Statistically significant: YES

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

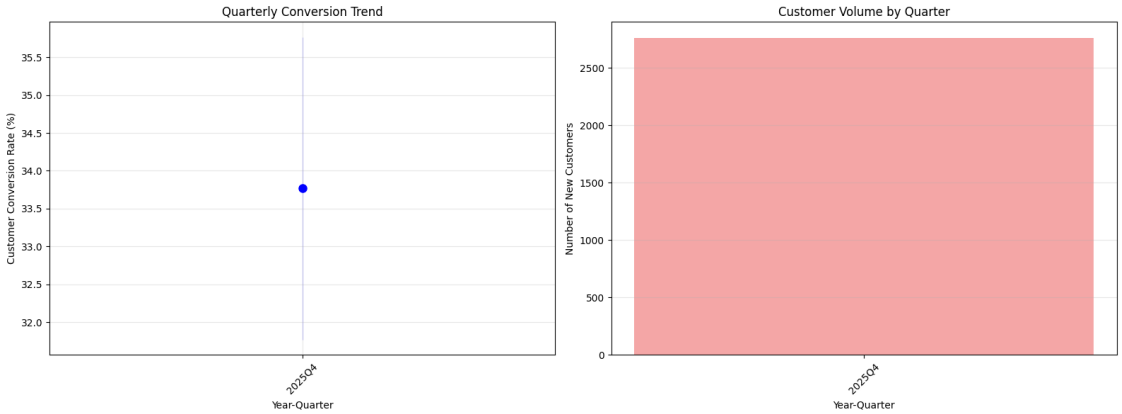
Top regions: ['Auvergne-Rhône-Alpes', 'Normandie', 'Hauts-de-France']
```




```
================================================================================
Year-over-year trends
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

Quarterly conversion rates:
                  mean  count
year_quarter
2025Q4        0.337676   2763
```

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================================================================================
Best and Worst Months Report
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
```

```
Median price: €6,251

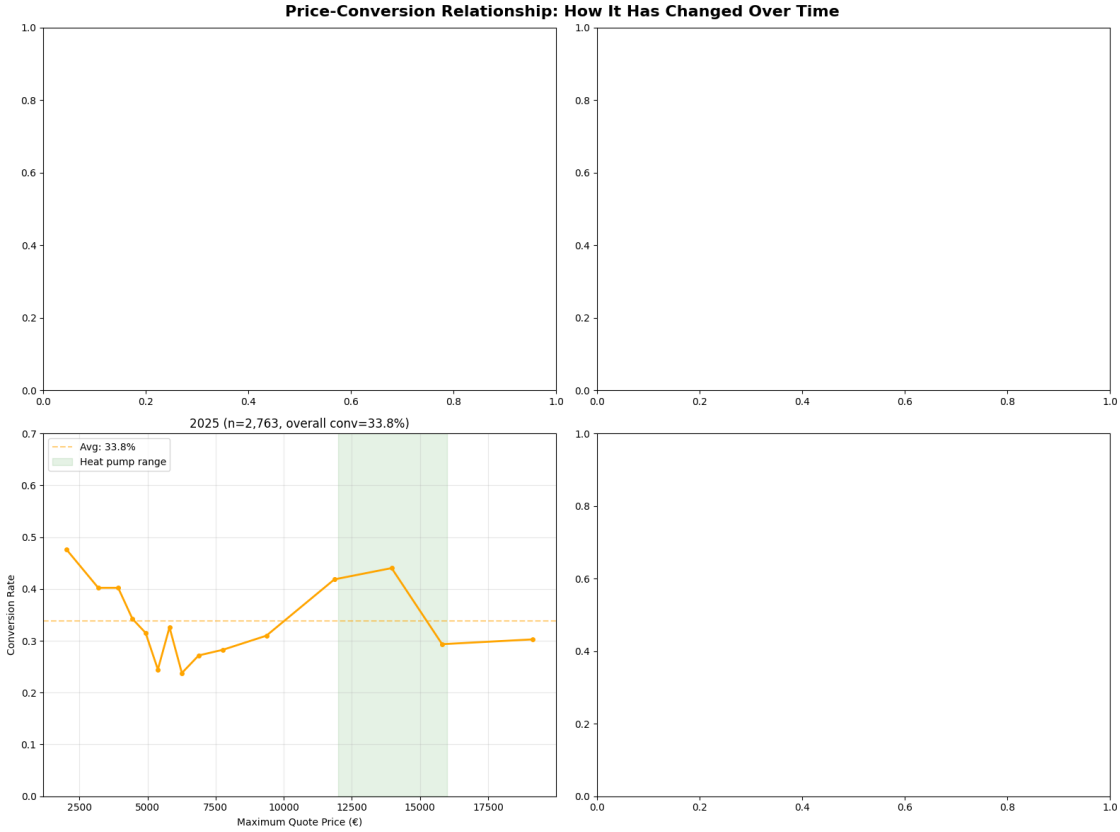
  Only one suspension period found: [False]
Skipping suspension comparison

Best month: 10 (37.0%, n=1076)

Best season: Fall (35.1%)
Worst season: Winter (30.7%)

Heat pump conversion by season:
  Winter: 41.0%
  Fall: 42.2%

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
```

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Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
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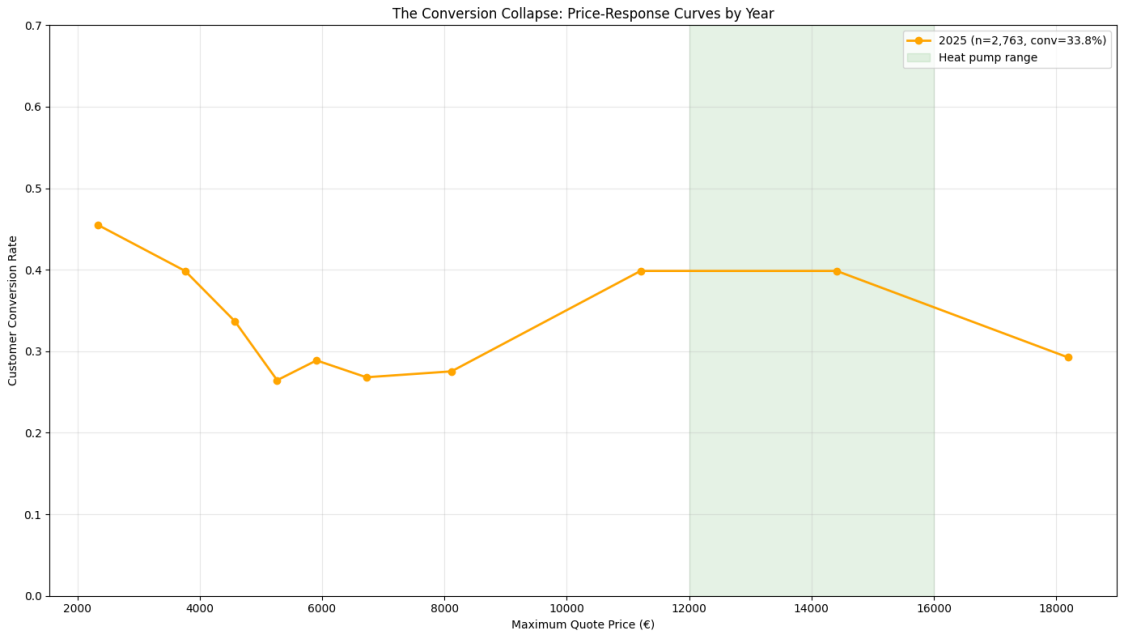


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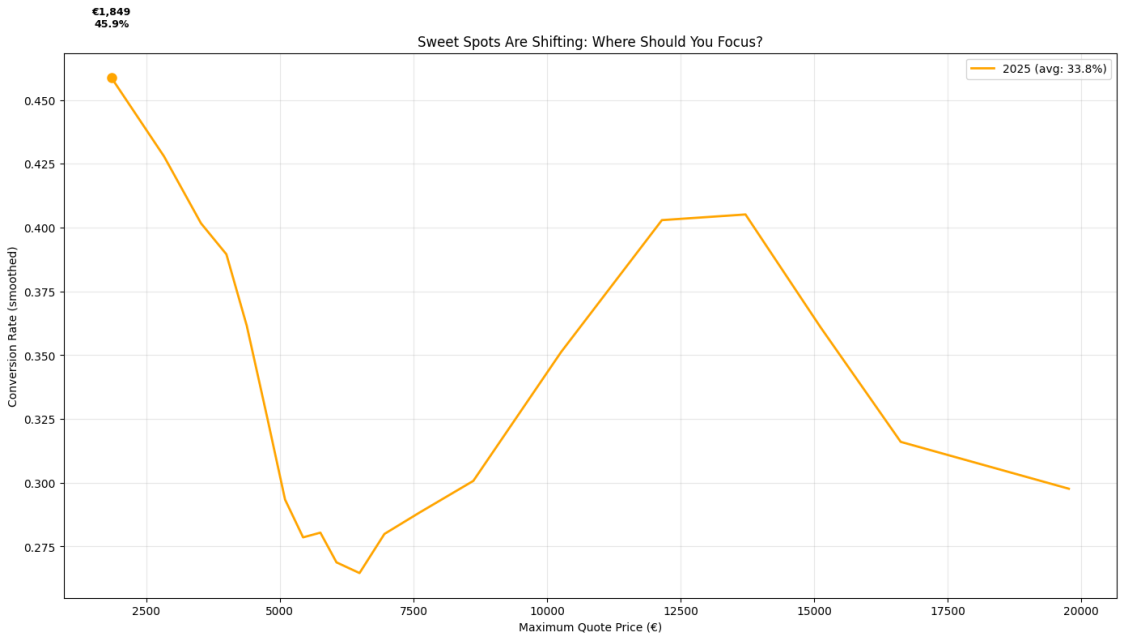
================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

Sweet spots by year (price points with conversion > avg + 3%):

2025 (overall avg: 33.8%):
  €1,849: 48.2% (n=139)
  €2,832: 42.8% (n=138)
  €3,999: 43.5% (n=138)
  €12,149: 43.5% (n=138)
  €13,714: 44.2% (n=138)
```

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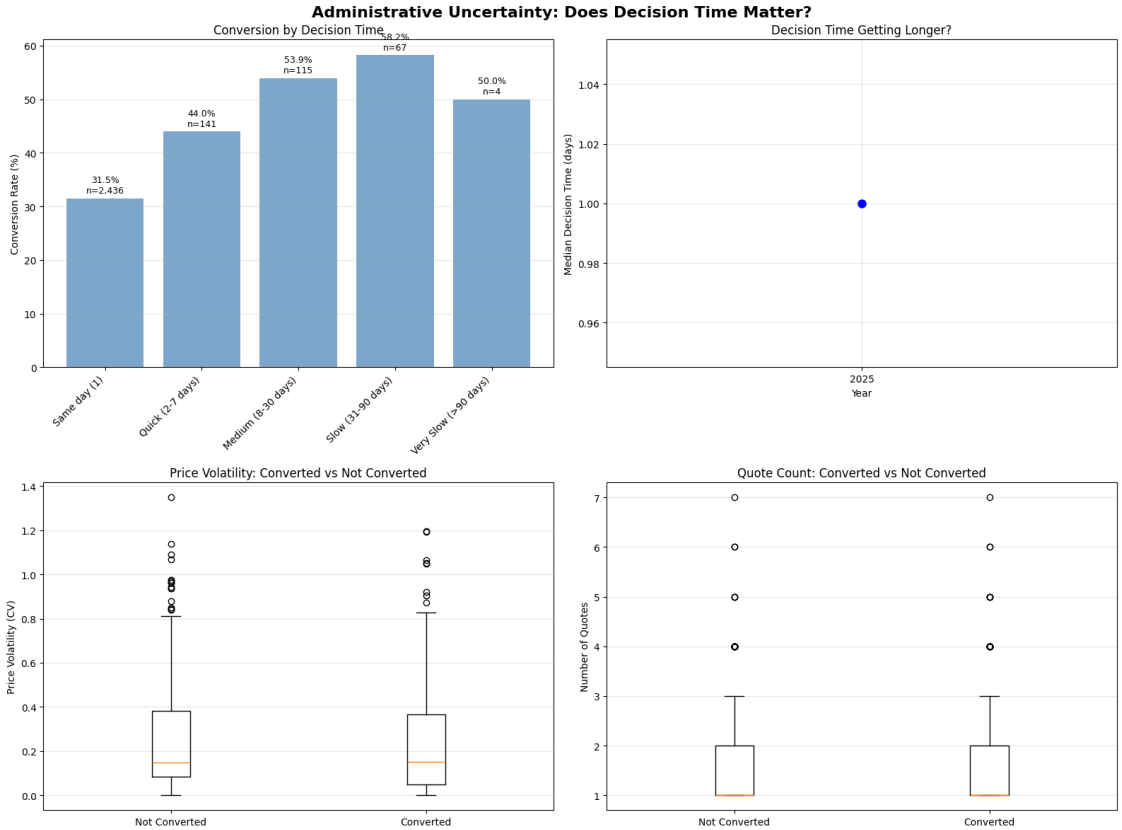
================================================================================
Administrative Uncertainty Effect
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

Decision time distribution:
count    2763.000000
mean        3.002895
std         8.847611
min         1.000000
25%         1.000000
50%         1.000000
75%         1.000000
max       106.000000
Name: decision_days, dtype: float64

Customers with 1-day decision: 2,436 (88.2%)
```

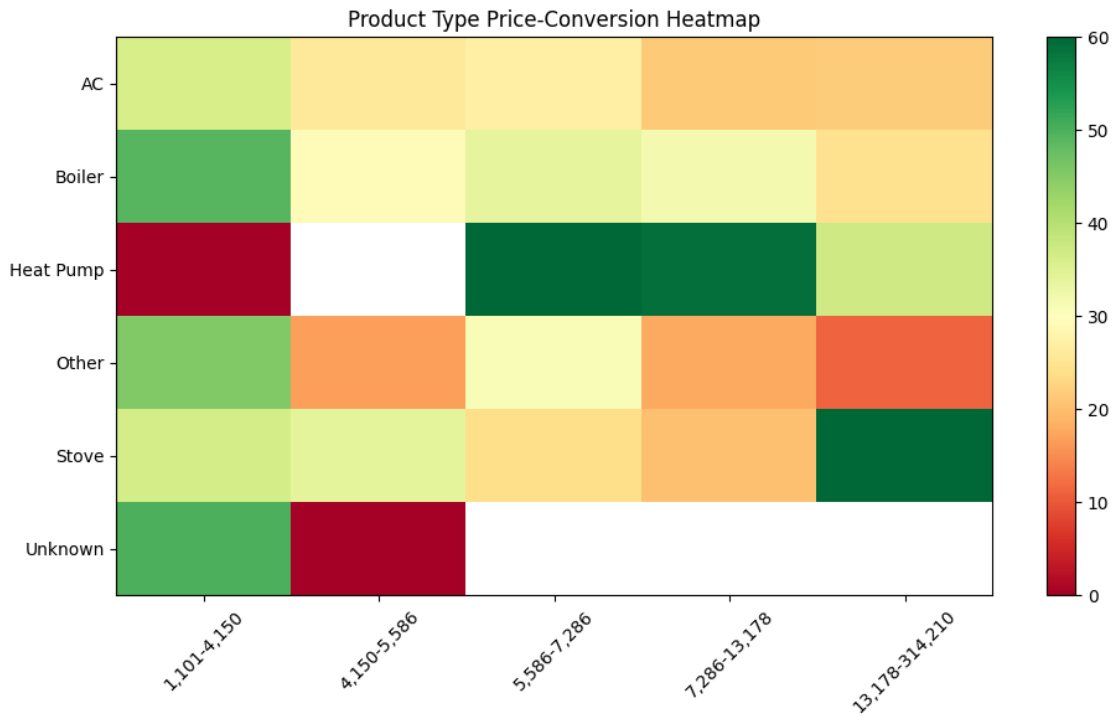



```
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1101.459, 4150.01]  (4150.01, 5586.29]  \
main_equipment_category
AC                                 36.153846          25.609756
Boiler                             49.019608          29.181495
Heat Pump                           0.000000                NaN
Other                              45.454545          16.666667
Stove                              36.363636          34.020619
Unknown                            50.000000           0.000000

price_bin_simple         (5586.29, 7286.95]  (7286.95, 13178.02]  \
main_equipment_category
AC                                26.966292          21.527778
Boiler                            33.870968          31.693989
Heat Pump                         60.000000          58.865248
Other                             30.769231          17.647059
Stove                             23.985240          20.253165
```

```
Unknown                                  NaN                   NaN

price_bin_simple         (13178.02, 314210.61]
main_equipment_category
AC                                  21.666667
Boiler                              24.528302
Heat Pump                           37.149533
Other                               11.111111
Stove                               60.000000
Unknown                                   NaN
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                          968
Poêle                              604
Pompe à chaleur                    531
Climatisation                      493
ECS : Chauffe-eau ou adoucisseur   128
Appareil hybride                    54
Photovoltaïque                      34
Autres                              21
```

```
Produit VMC                          12
Emetteur de chauffage  ou chappe     11
Plomberie Sanitaire                   8
Unknown                               3
Name: count, dtype: int64

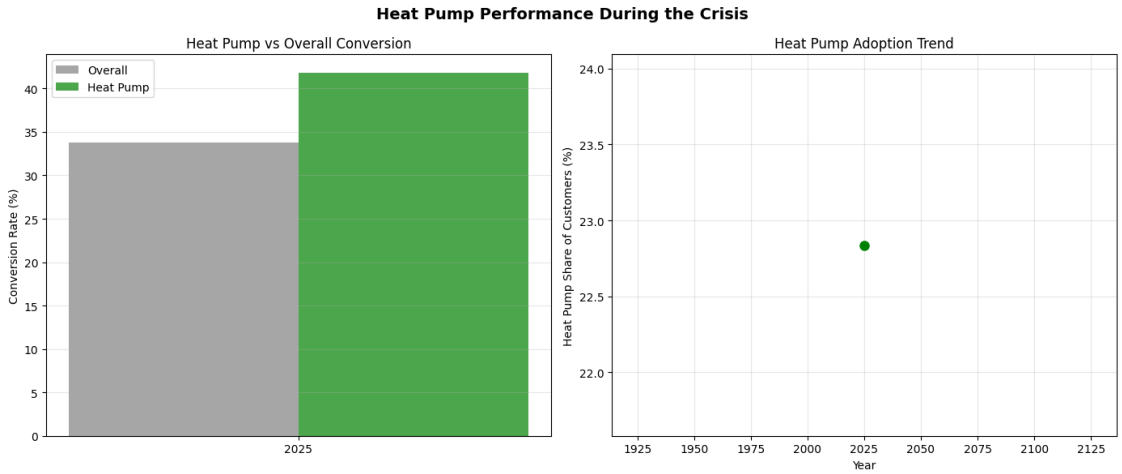
Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251
```

```
================================================================================
Heat Pump Performance Over Time
================================================================================

Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
period
```

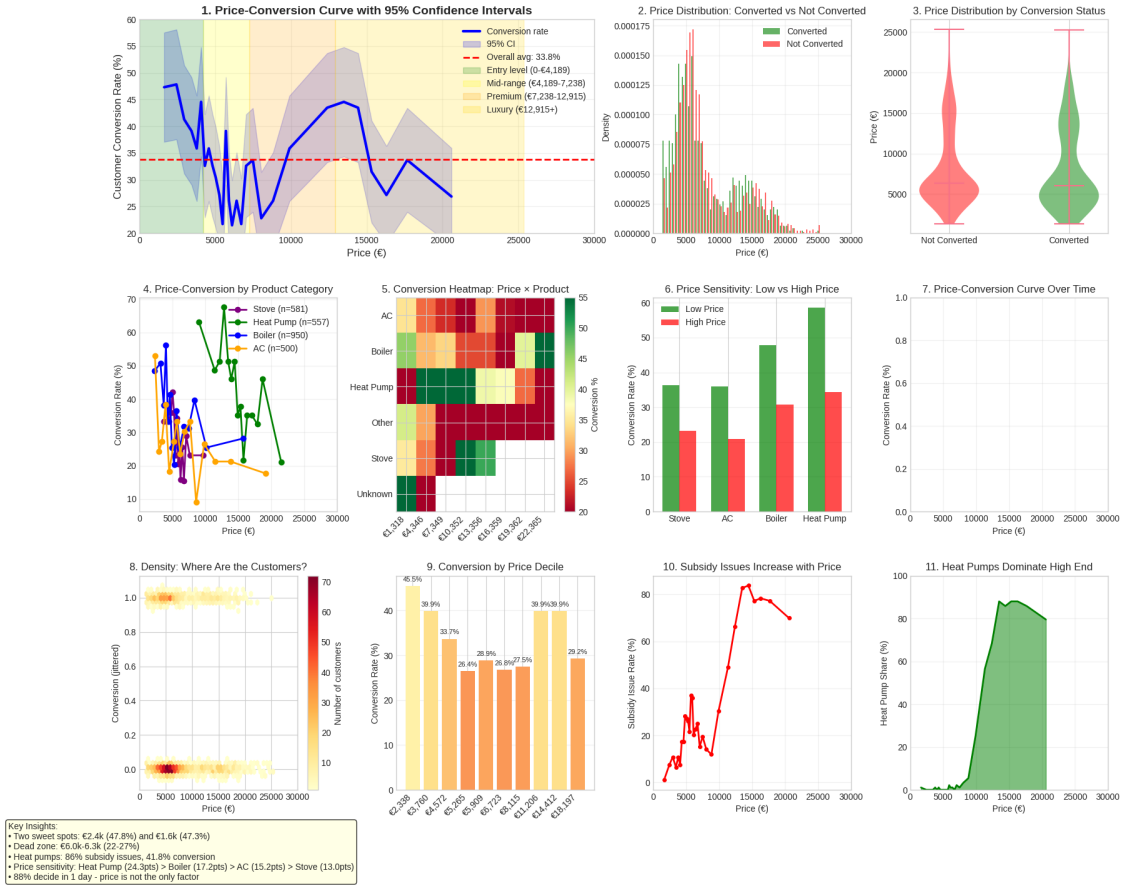
```
2025                0.418              631  14952.297                0.861
```

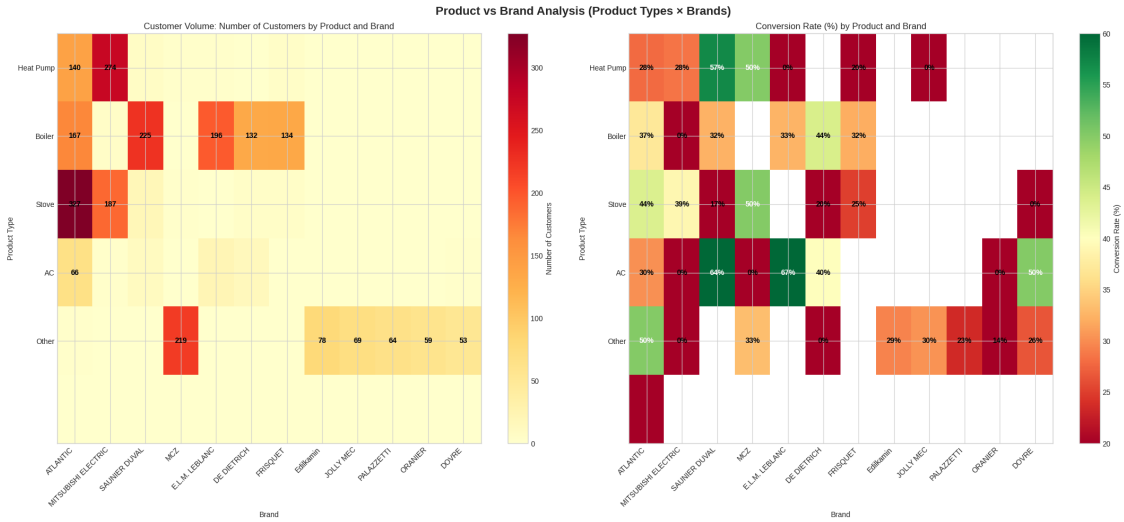



```
Analyzing 2,763 customers
Price range: €1,343 to €25,369
Median price: €6,251

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €2.4k (47.8%) and €1.6k (47.3%)
Dead zone: €6.0k-6.3k (22-27%)
Heat pumps: 86% subsidy issues, 41.8% conversion
Price sensitivity: Heat Pump (24.3pts) > Boiler (17.2pts) > AC (15.2pts) > Stove
(13.0pts)
Same day decisions: 88%
```

Price-Conversion Dependency Dashboard





```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  SAUNIER DUVAL  MCZ
E.L.M. LEBLANC  DE DIETRICH  FRISQUET  Edilkamin  JOLLY MEC  PALAZZETTI  ORANIER
DOVRE
main_equipment_category
AC                            140                  274              7    2
2            0         5          0          1           0        0      0
Boiler                        167                    4            225    0
196          132       134          0          0           0        0      0
Heat Pump                     327                  187             18    2
0            5         4          0          0           0        0      1
Other                          66                    2             11    1
21           15         0          0          0           0        1      2
Stove                           2                    1              0  219
0            1         0         78         69          64       59     53
Unknown                         1                    0              0    0
0            0         0          0          0           0        0      0

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  SAUNIER DUVAL   MCZ
E.L.M. LEBLANC  DE DIETRICH  FRISQUET  Edilkamin  JOLLY MEC  PALAZZETTI  ORANIER
DOVRE
main_equipment_category
AC                           27.9                 28.5           57.1  50.0
0.0          NaN      20.0        NaN        0.0         NaN      NaN    NaN
Boiler                       37.1                  0.0           32.4   NaN
32.7         43.9      32.1        NaN        NaN         NaN      NaN    NaN
Heat Pump                    43.7                 39.0           16.7  50.0
NaN         20.0      25.0        NaN        NaN         NaN      NaN    0.0
Other                        30.3                  0.0           63.6   0.0
66.7         40.0       NaN        NaN        NaN         NaN      0.0   50.0
Stove                        50.0                  0.0            NaN  33.3
NaN          0.0       NaN       29.5       30.4        23.4     13.6   26.4
Unknown                       0.0                  NaN            NaN   NaN
NaN          NaN       NaN        NaN        NaN         NaN      NaN    NaN

================================================================================
  BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------
```

```
Heat Pump:
  Best brand: MCZ
  Conversion: 50.0%
  Customers: 2
  Top 3 brands:
    1. MCZ: 50.0% (n=2)
    2. ATLANTIC: 43.7% (n=327)
    3. MITSUBISHI ELECTRIC: 39.0% (n=187)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 43.9%
  Customers: 132
  Top 3 brands:
    1. DE DIETRICH: 43.9% (n=132)
    2. ATLANTIC: 37.1% (n=167)
    3. E.L.M. LEBLANC: 32.7% (n=196)

Stove:
  Best brand: ATLANTIC
  Conversion: 50.0%
  Customers: 2
  Top 3 brands:
    1. ATLANTIC: 50.0% (n=2)
    2. MCZ: 33.3% (n=219)
    3. JOLLY MEC: 30.4% (n=69)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 57.1%
  Customers: 7
  Top 3 brands:
    1. SAUNIER DUVAL: 57.1% (n=7)
    2. MCZ: 50.0% (n=2)
    3. MITSUBISHI ELECTRIC: 28.5% (n=274)

Other:
  Best brand: E.L.M. LEBLANC
  Conversion: 66.7%
  Customers: 21
  Top 3 brands:
    1. E.L.M. LEBLANC: 66.7% (n=21)
    2. SAUNIER DUVAL: 63.6% (n=11)
    3. DOVRE: 50.0% (n=2)

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------
```

```
ATLANTIC:
  Best product: Stove
  Conversion: 50.0%
  Customers: 2
  All products:
    Stove: 50.0% (n=2)
    Heat Pump: 43.7% (n=327)
    Boiler: 37.1% (n=167)
    Other: 30.3% (n=66)
    AC: 27.9% (n=140)
    Unknown: 0.0% (n=1)

MITSUBISHI ELECTRIC:
  Best product: Heat Pump
  Conversion: 39.0%
  Customers: 187
  All products:
    Heat Pump: 39.0% (n=187)
    AC: 28.5% (n=274)
    Boiler: 0.0% (n=4)
    Other: 0.0% (n=2)
    Stove: 0.0% (n=1)

SAUNIER DUVAL:
  Best product: Other
  Conversion: 63.6%
  Customers: 11
  All products:
    Other: 63.6% (n=11)
    AC: 57.1% (n=7)
    Boiler: 32.4% (n=225)
    Heat Pump: 16.7% (n=18)

MCZ:
  Best product: AC
  Conversion: 50.0%
  Customers: 2
  All products:
    AC: 50.0% (n=2)
    Heat Pump: 50.0% (n=2)
    Stove: 33.3% (n=219)
    Other: 0.0% (n=1)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 66.7%
  Customers: 21
```

```
  All products:
    Other: 66.7% (n=21)
    Boiler: 32.7% (n=196)
    AC: 0.0% (n=2)

DE DIETRICH:
  Best product: Boiler
  Conversion: 43.9%
  Customers: 132
  All products:
    Boiler: 43.9% (n=132)
    Other: 40.0% (n=15)
    Heat Pump: 20.0% (n=5)
    Stove: 0.0% (n=1)

FRISQUET:
  Best product: Boiler
  Conversion: 32.1%
  Customers: 134
  All products:
    Boiler: 32.1% (n=134)
    Heat Pump: 25.0% (n=4)
    AC: 20.0% (n=5)

Edilkamin:
  Best product: Stove
  Conversion: 29.5%
  Customers: 78
  All products:
    Stove: 29.5% (n=78)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

Heat Pump:
  Volume leader: ATLANTIC (327 customers, 60.1% share)
  Conversion leader: MCZ (50.0%)

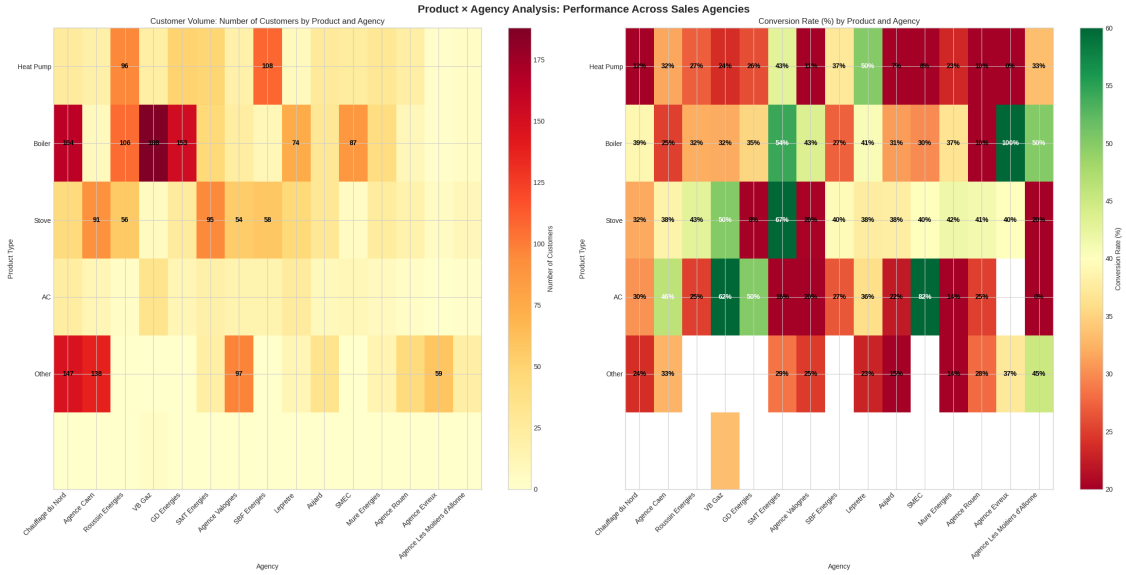
Boiler:
  Volume leader: SAUNIER DUVAL (225 customers, 26.2% share)
  Conversion leader: DE DIETRICH (43.9%)

Stove:
  Volume leader: MCZ (219 customers, 40.1% share)
  Conversion leader: ATLANTIC (50.0%)
```

```
AC:
  Volume leader: MITSUBISHI ELECTRIC (274 customers, 63.6% share)
  Conversion leader: SAUNIER DUVAL (57.1%)

Other:
  Volume leader: ATLANTIC (66 customers, 55.5% share)
  Conversion leader: E.L.M. LEBLANC (66.7%)

Top 15 agencies by customer volume:
  Chauffage du Nord: 403 customers, 30.5% conversion
  Agence Caen: 269 customers, 34.9% conversion
  Roussin Energies: 262 customers, 32.4% conversion
  VB Gaz: 252 customers, 35.7% conversion
  GD Energies: 240 customers, 31.2% conversion
  SMT Energies: 230 customers, 51.7% conversion
  Agence Valognes: 207 customers, 24.2% conversion
  SBF Energies: 192 customers, 36.5% conversion
  Lepretre: 165 customers, 38.2% conversion
  Aujard: 128 customers, 22.7% conversion
  SMEC: 127 customers, 30.7% conversion
  Mure Energies: 116 customers, 30.2% conversion
  Agence Rouen: 95 customers, 24.2% conversion
  Agence Evreux: 66 customers, 37.9% conversion
  Agence Les Moitiers d'Allonne: 39 customers, 35.9% conversion
```




```
================================================================================
PRODUCT × AGENCY ANALYSIS
```

```
================================================================================

  CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Agence Caen  Roussin Energies  VB
Gaz  GD Energies  SMT Energies  Agence Valognes  SBF Energies  Lepretre  Aujard
SMEC  Mure Energies  Agence Rouen  Agence Evreux  Agence Les Moitiers d'Allonne
main_equipment_category
AC                                      25           19                96
21           50            49               18           108         8      27
24             30            21              1                                3
Boiler                                 164            8               106
188          153            46               23            11        74      29
87             41            10              1                                4
Heat Pump                               44           91                56
6           25            95               54            58        45      29
5             24            17              5                               10
Other                                   23           13                 4
34           12            19               15            15        25       9
11              7             4              0                                2
Stove                                  147          138                 0
0            0            21               97             0        13      34
0             14            43             59                               20
Unknown                                  0            0                 0
3            0             0                0             0         0       0
0              0             0              0                                0

================================================================================
  CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Agence Caen  Roussin Energies  VB
Gaz  GD Energies  SMT Energies  Agence Valognes  SBF Energies  Lepretre  Aujard
SMEC  Mure Energies  Agence Rouen  Agence Evreux  Agence Les Moitiers d'Allonne
main_equipment_category
AC                                    12.0         31.6              27.1
23.8         26.0          42.9             11.1          37.0      50.0     7.4
8.3           23.3           9.5            0.0                            33.3
Boiler                                39.0         25.0              32.1
31.9         35.3          54.3             43.5          27.3      40.5    31.0
29.9           36.6          10.0          100.0                           50.0
Heat Pump                             31.8         38.5              42.9
50.0          8.0          67.4             20.4          39.7      37.8    37.9
40.0           41.7          41.2           40.0                           20.0
Other                                 30.4         46.2              25.0
61.8         50.0          15.8             20.0          26.7      36.0    22.2
81.8           14.3          25.0            NaN                            0.0
```

```
Stove                                   23.8         32.6                NaN
NaN          NaN          28.6             24.7           NaN      23.1    14.7
NaN           14.3          27.9           37.3                            45.0
Unknown                                  NaN          NaN                NaN
33.3          NaN           NaN              NaN           NaN       NaN     NaN
NaN            NaN           NaN            NaN                             NaN

================================================================================
  BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------

Heat Pump:
  Best agency: SMT Energies
  Conversion: 67.4%
  Customers: 95
  Top 3 agencies:
    1. SMT Energies: 67.4% (n=95)
    2. VB Gaz: 50.0% (n=6)
    3. Roussin Energies: 42.9% (n=56)

Boiler:
  Best agency: Agence Evreux
  Conversion: 100.0%
  Customers: 1
  Top 3 agencies:
    1. Agence Evreux: 100.0% (n=1)
    2. SMT Energies: 54.3% (n=46)
    3. Agence Les Moitiers d'Allonne: 50.0% (n=4)

Stove:
  Best agency: Agence Les Moitiers d'Allonne
  Conversion: 45.0%
  Customers: 20
  Top 3 agencies:
    1. Agence Les Moitiers d'Allonne: 45.0% (n=20)
    2. Agence Evreux: 37.3% (n=59)
    3. Agence Caen: 32.6% (n=138)

AC:
  Best agency: Lepretre
  Conversion: 50.0%
  Customers: 8
  Top 3 agencies:
    1. Lepretre: 50.0% (n=8)
    2. SMT Energies: 42.9% (n=49)
    3. SBF Energies: 37.0% (n=108)
```

```
Other:
  Best agency: SMEC
  Conversion: 81.8%
  Customers: 11
  Top 3 agencies:
    1. SMEC: 81.8% (n=11)
    2. VB Gaz: 61.8% (n=34)
    3. GD Energies: 50.0% (n=12)

================================================================================
 BEST PRODUCT BY AGENCY
--------------------------------------------------------------------------------
--------------------

Chauffage du Nord:
  Best product: Boiler
  Conversion: 39.0%
  Customers: 164
  All products:
    Boiler: 39.0% (n=164)
    Heat Pump: 31.8% (n=44)
    Other: 30.4% (n=23)
    Stove: 23.8% (n=147)
    AC: 12.0% (n=25)

Agence Caen:
  Best product: Other
  Conversion: 46.2%
  Customers: 13
  All products:
    Other: 46.2% (n=13)
    Heat Pump: 38.5% (n=91)
    Stove: 32.6% (n=138)
    AC: 31.6% (n=19)
    Boiler: 25.0% (n=8)

Roussin Energies:
  Best product: Heat Pump
  Conversion: 42.9%
  Customers: 56
  All products:
    Heat Pump: 42.9% (n=56)
    Boiler: 32.1% (n=106)
    AC: 27.1% (n=96)
    Other: 25.0% (n=4)

VB Gaz:
  Best product: Other
```

```
  Conversion: 61.8%
  Customers: 34
  All products:
    Other: 61.8% (n=34)
    Heat Pump: 50.0% (n=6)
    Unknown: 33.3% (n=3)
    Boiler: 31.9% (n=188)
    AC: 23.8% (n=21)

GD Energies:
  Best product: Other
  Conversion: 50.0%
  Customers: 12
  All products:
    Other: 50.0% (n=12)
    Boiler: 35.3% (n=153)
    AC: 26.0% (n=50)
    Heat Pump: 8.0% (n=25)

SMT Energies:
  Best product: Heat Pump
  Conversion: 67.4%
  Customers: 95
  All products:
    Heat Pump: 67.4% (n=95)
    Boiler: 54.3% (n=46)
    AC: 42.9% (n=49)
    Stove: 28.6% (n=21)
    Other: 15.8% (n=19)

Agence Valognes:
  Best product: Boiler
  Conversion: 43.5%
  Customers: 23
  All products:
    Boiler: 43.5% (n=23)
    Stove: 24.7% (n=97)
    Heat Pump: 20.4% (n=54)
    Other: 20.0% (n=15)
    AC: 11.1% (n=18)

SBF Energies:
  Best product: Heat Pump
  Conversion: 39.7%
  Customers: 58
  All products:
    Heat Pump: 39.7% (n=58)
    AC: 37.0% (n=108)
```

```
    Boiler: 27.3% (n=11)
    Other: 26.7% (n=15)

Lepretre:
  Best product: AC
  Conversion: 50.0%
  Customers: 8
  All products:
    AC: 50.0% (n=8)
    Boiler: 40.5% (n=74)
    Heat Pump: 37.8% (n=45)
    Other: 36.0% (n=25)
    Stove: 23.1% (n=13)

Aujard:
  Best product: Heat Pump
  Conversion: 37.9%
  Customers: 29
  All products:
    Heat Pump: 37.9% (n=29)
    Boiler: 31.0% (n=29)
    Other: 22.2% (n=9)
    Stove: 14.7% (n=34)
    AC: 7.4% (n=27)

================================================================================
  AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                              conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
Chauffage du Nord                        30.5             403           12.2
44.2   6302.186
Agence Caen                              34.9             269           35.7
49.8   9376.482
Roussin Energies                         32.4             262           23.7
22.5   8996.472
VB Gaz                                   35.7             252            3.2
3.6   5408.402
GD Energies                              31.2             240           15.4
13.8   6273.609
SMT Energies                             51.7             230           45.7
57.8   8103.644
Agence Valognes                          24.2             207           31.4
46.4   8916.642
SBF Energies                             36.5             192           33.9
```

```
28.6  11748.518
Lepretre                                38.2            165          28.5
33.9   7982.366
Aujard                                  22.7            128          23.4
21.1   8269.907
SMEC                                    30.7            127           5.5
10.2   6024.826
Mure Energies                           30.2            116          24.1
25.0   7865.042
Agence Rouen                            24.2             95          21.1
46.3   8906.089
Agence Evreux                           37.9             66           7.6
31.8   6859.633
Agence Les Moitiers d'Allonne           35.9             39          25.6
41.0   8640.483

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Heat Pump:
  Conversion leader: SMT Energies (67.4%, n=95)

Boiler:
  Conversion leader: Agence Evreux (100.0%, n=1)

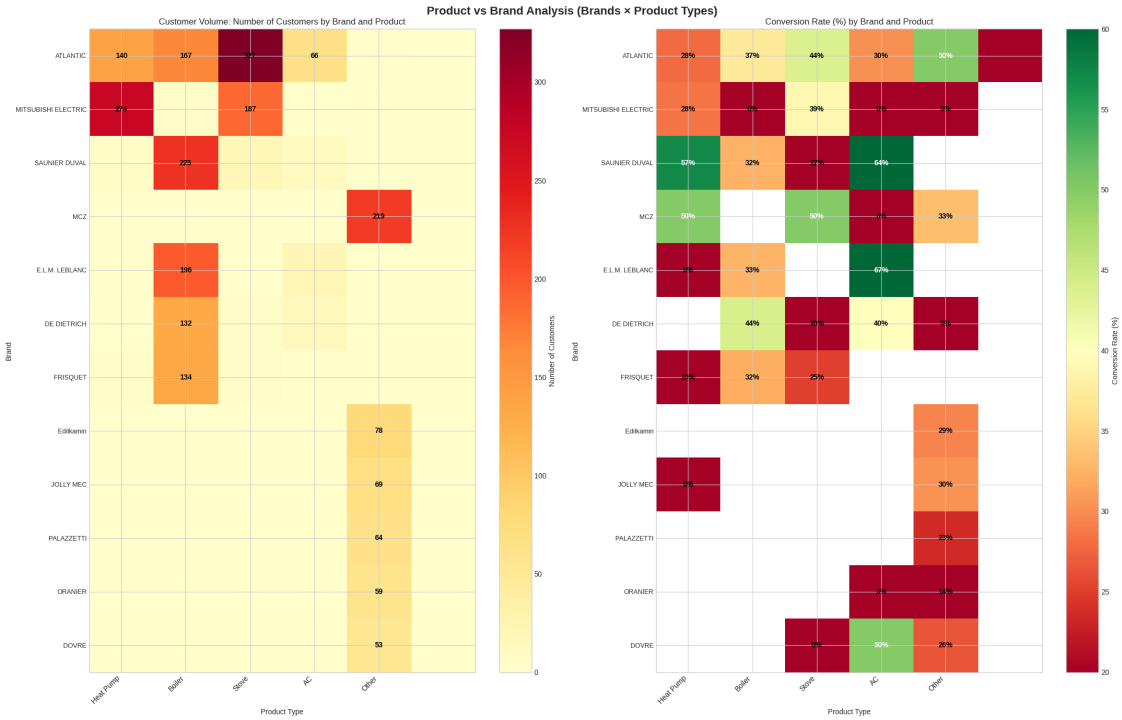
Stove:
  Conversion leader: Agence Les Moitiers d'Allonne (45.0%, n=20)

AC:
  Conversion leader: Lepretre (50.0%, n=8)

Other:
  Conversion leader: SMEC (81.8%, n=11)

================================================================================
 HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  SMT Energies: 67.4% (n=95)
  VB Gaz: 50.0% (n=6)
  Roussin Energies: 42.9% (n=56)
  Mure Energies: 41.7% (n=24)
  Agence Rouen: 41.2% (n=17)
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category   AC  Boiler  Heat Pump  Other  Stove  Unknown
main_brand
ATLANTIC                 140     167        327     66      2        1
MITSUBISHI ELECTRIC      274       4        187      2      1        0
SAUNIER DUVAL              7     225         18     11      0        0
MCZ                        2       0          2      1    219        0
E.L.M. LEBLANC             2     196          0     21      0        0
DE DIETRICH                0     132          5     15      1        0
FRISQUET                   5     134          4      0      0        0
Edilkamin                  0       0          0      0     78        0
JOLLY MEC                  1       0          0      0     69        0
PALAZZETTI                 0       0          0      0     64        0
ORANIER                    0       0          0      1     59        0
DOVRE                      0       0          1      2     53        0

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove  Unknown
main_brand
ATLANTIC                 27.9    37.1       43.7   30.3   50.0      0.0
MITSUBISHI ELECTRIC      28.5     0.0       39.0    0.0    0.0      NaN
SAUNIER DUVAL            57.1    32.4       16.7   63.6    NaN      NaN
MCZ                      50.0     NaN       50.0    0.0   33.3      NaN
E.L.M. LEBLANC            0.0    32.7        NaN   66.7    NaN      NaN
DE DIETRICH               NaN    43.9       20.0   40.0    0.0      NaN
FRISQUET                 20.0    32.1       25.0    NaN    NaN      NaN
Edilkamin                 NaN     NaN        NaN    NaN   29.5      NaN
JOLLY MEC                 0.0     NaN        NaN    NaN   30.4      NaN
PALAZZETTI                NaN     NaN        NaN    NaN   23.4      NaN
ORANIER                   NaN     NaN        NaN    0.0   13.6      NaN
DOVRE                     NaN     NaN        0.0   50.0   26.4      NaN

================================================================================
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Stove
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. Stove: 50.0% (n=2)
    2. Heat Pump: 43.7% (n=327)
    3. Boiler: 37.1% (n=167)

MITSUBISHI ELECTRIC:
  Best product: Heat Pump
  Conversion: 39.0%
  Customers: 187
  Top 3 products:
    1. Heat Pump: 39.0% (n=187)
    2. AC: 28.5% (n=274)
    3. Boiler: 0.0% (n=4)

SAUNIER DUVAL:
  Best product: Other
  Conversion: 63.6%
  Customers: 11
  Top 3 products:
    1. Other: 63.6% (n=11)
    2. AC: 57.1% (n=7)
    3. Boiler: 32.4% (n=225)

MCZ:
```

```
  Best product: AC
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. AC: 50.0% (n=2)
    2. Heat Pump: 50.0% (n=2)
    3. Stove: 33.3% (n=219)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 66.7%
  Customers: 21
  Top 3 products:
    1. Other: 66.7% (n=21)
    2. Boiler: 32.7% (n=196)
    3. AC: 0.0% (n=2)

DE DIETRICH:
  Best product: Boiler
  Conversion: 43.9%
  Customers: 132
  Top 3 products:
    1. Boiler: 43.9% (n=132)
    2. Other: 40.0% (n=15)
    3. Heat Pump: 20.0% (n=5)

FRISQUET:
  Best product: Boiler
  Conversion: 32.1%
  Customers: 134
  Top 3 products:
    1. Boiler: 32.1% (n=134)
    2. Heat Pump: 25.0% (n=4)
    3. AC: 20.0% (n=5)

Edilkamin:
  Best product: Stove
  Conversion: 29.5%
  Customers: 78
  Top 3 products:
    1. Stove: 29.5% (n=78)

JOLLY MEC:
  Best product: Stove
  Conversion: 30.4%
  Customers: 69
  Top 3 products:
    1. Stove: 30.4% (n=69)
```

```
    2. AC: 0.0% (n=1)

PALAZZETTI:
  Best product: Stove
  Conversion: 23.4%
  Customers: 64
  Top 3 products:
    1. Stove: 23.4% (n=64)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: MCZ
  Conversion: 50.0%
  Customers: 2
  Top 5 brands:
    1. MCZ: 50.0% (n=2)
    2. ATLANTIC: 43.7% (n=327)
    3. MITSUBISHI ELECTRIC: 39.0% (n=187)
    4. FRISQUET: 25.0% (n=4)
    5. DE DIETRICH: 20.0% (n=5)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 43.9%
  Customers: 132
  Top 5 brands:
    1. DE DIETRICH: 43.9% (n=132)
    2. ATLANTIC: 37.1% (n=167)
    3. E.L.M. LEBLANC: 32.7% (n=196)
    4. SAUNIER DUVAL: 32.4% (n=225)
    5. FRISQUET: 32.1% (n=134)

Stove:
  Best brand: ATLANTIC
  Conversion: 50.0%
  Customers: 2
  Top 5 brands:
    1. ATLANTIC: 50.0% (n=2)
    2. MCZ: 33.3% (n=219)
    3. JOLLY MEC: 30.4% (n=69)
    4. Edilkamin: 29.5% (n=78)
    5. DOVRE: 26.4% (n=53)

AC:
  Best brand: SAUNIER DUVAL
```

```
  Conversion: 57.1%
  Customers: 7
  Top 5 brands:
    1. SAUNIER DUVAL: 57.1% (n=7)
    2. MCZ: 50.0% (n=2)
    3. MITSUBISHI ELECTRIC: 28.5% (n=274)
    4. ATLANTIC: 27.9% (n=140)
    5. FRISQUET: 20.0% (n=5)

Other:
  Best brand: E.L.M. LEBLANC
  Conversion: 66.7%
  Customers: 21
  Top 5 brands:
    1. E.L.M. LEBLANC: 66.7% (n=21)
    2. SAUNIER DUVAL: 63.6% (n=11)
    3. DOVRE: 50.0% (n=2)
    4. DE DIETRICH: 40.0% (n=15)
    5. ATLANTIC: 30.3% (n=66)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Stove: 50.0% (n=2)
  Heat Pump: 43.7% (n=327)
  Boiler: 37.1% (n=167)

MITSUBISHI ELECTRIC:
  Heat Pump: 39.0% (n=187)
  AC: 28.5% (n=274)
  Boiler: 0.0% (n=4)

SAUNIER DUVAL:
  Other: 63.6% (n=11)
  AC: 57.1% (n=7)
  Boiler: 32.4% (n=225)

MCZ:
  AC: 50.0% (n=2)
  Heat Pump: 50.0% (n=2)
  Stove: 33.3% (n=219)

E.L.M. LEBLANC:
  Other: 66.7% (n=21)
```

```
  Boiler: 32.7% (n=196)
  AC: 0.0% (n=2)

DE DIETRICH:
  Boiler: 43.9% (n=132)
  Other: 40.0% (n=15)
  Heat Pump: 20.0% (n=5)

FRISQUET:
  Boiler: 32.1% (n=134)
  Heat Pump: 25.0% (n=4)
  AC: 20.0% (n=5)

Edilkamin:
  Stove: 29.5% (n=78)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Heat Pump:
  Volume leader: ATLANTIC (327 customers, 60.1% share)
  Conversion leader: MCZ (50.0%)

Boiler:
  Volume leader: SAUNIER DUVAL (225 customers, 26.2% share)
  Conversion leader: DE DIETRICH (43.9%)

Stove:
  Volume leader: MCZ (219 customers, 40.1% share)
  Conversion leader: ATLANTIC (50.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (274 customers, 63.6% share)
  Conversion leader: SAUNIER DUVAL (57.1%)

Other:
  Volume leader: ATLANTIC (66 customers, 55.5% share)
  Conversion leader: E.L.M. LEBLANC (66.7%)

 (main_equipment_category   AC  Boiler  Heat Pump  Other  Stove  Unknown
  main_brand
  ATLANTIC                 140     167        327     66      2        1
  MITSUBISHI ELECTRIC      274       4        187      2      1        0
  SAUNIER DUVAL              7     225         18     11      0        0
  MCZ                        2       0          2      1    219        0
  E.L.M. LEBLANC             2     196          0     21      0        0
```

```
DE DIETRICH                0     132          5     15      1        0
FRISQUET                   5     134          4      0      0        0
Edilkamin                  0       0          0      0     78        0
JOLLY MEC                  1       0          0      0     69        0
PALAZZETTI                 0       0          0      0     64        0
ORANIER                    0       0          0      1     59        0
DOVRE                      0       0          1      2     53        0,
main_equipment_category         AC     Boiler  Heat Pump      Other  \
main_brand
ATLANTIC                 27.857143  37.125749  43.730887  30.303030
MITSUBISHI ELECTRIC      28.467153   0.000000  39.037433   0.000000
SAUNIER DUVAL            57.142857  32.444444  16.666667  63.636364
MCZ                      50.000000        NaN  50.000000   0.000000
E.L.M. LEBLANC            0.000000  32.653061        NaN  66.666667
DE DIETRICH                    NaN  43.939394  20.000000  40.000000
FRISQUET                 20.000000  32.089552  25.000000        NaN
Edilkamin                      NaN        NaN        NaN        NaN
JOLLY MEC                 0.000000        NaN        NaN        NaN
PALAZZETTI                     NaN        NaN        NaN        NaN
ORANIER                        NaN        NaN        NaN   0.000000
DOVRE                          NaN        NaN   0.000000  50.000000

main_equipment_category      Stove  Unknown
main_brand
ATLANTIC                 50.000000      0.0
MITSUBISHI ELECTRIC       0.000000      NaN
SAUNIER DUVAL                  NaN      NaN
MCZ                      33.333333      NaN
E.L.M. LEBLANC                 NaN      NaN
DE DIETRICH               0.000000      NaN
FRISQUET                       NaN      NaN
Edilkamin                29.487179      NaN
JOLLY MEC                30.434783      NaN
PALAZZETTI               23.437500      NaN
ORANIER                  13.559322      NaN
DOVRE                    26.415094      NaN  )
```

```python
import ipynbname
import subprocess
import os
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
```

```
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False
```

```python
    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
 Notebook: _eda_price_2025_Q4.ipynb
 PDF path: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025_Q4.pdf
 Converting to PDF…
 PDF saved: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025_Q4.pdf
```